Etude de la synchronisation des oscillateurs

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option informatique

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Introduction



Figure 1 – Dispositif permettant d'observer la synchronisation de deux métronomes

Plan de l'exposé

1. Le modèle de Kuramoto
2. Approche numérique
3. Approche analytique
4. Conclusion
5. Annexe

- Equation de la phase d'un oscillateur sans couplage $i \in [\![1, N]\!]$:

$$\theta_i(t) = \omega_i t + \varphi_i^0 \text{ avec } \omega_i = \dot{\theta}_i(t = 0)$$

$$\text{donc } \dot{\theta}_i(t) = \omega_i$$

- Equation de la phase avec couplage :

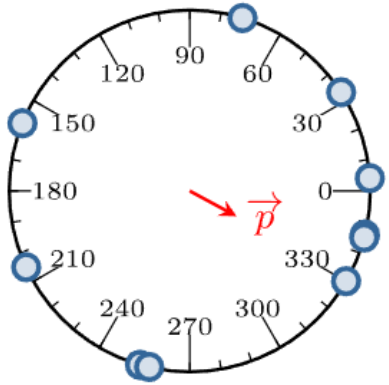
$$\dot{\theta}_i(t) = \omega_i + \sum_{k \neq j} K_{jk}(t)$$

- Equation de la phase avec le coefficient de couplage de Kuramoto (K) :

$$\dot{\theta}_i(t) = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j(t) - \theta_i(t))$$

- Expression du paramètre d'ordre

$$\vec{p} = r(t)\,\mathrm{e}^{\mathrm{i}\psi(t)} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{e}^{\mathrm{i}\theta_j(t)}$$



- Réécriture de l'expression de la phase

$$r(t)\,\mathrm{e}^{\mathrm{i}(\psi(t)-\theta_i(t))} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{e}^{\mathrm{i}(\theta_j(t)-\theta_i(t))}$$

L'équation de la phase devient : $\boxed{\dot{\theta}_i(t) = \omega_i + Kr(t)\sin(\psi(t) - \theta_i(t))}$

$$\frac{\mathrm{d}\theta_i(t)}{\mathrm{d}t} = \omega_i + Kr(t)\sin(\psi(t) - \theta_i(t))$$

donc entre t et $t + \mathrm{d}t$, $\theta_i(t+\mathrm{d}t) \approx \theta_i(t) + (\omega_i + Kr(t)\sin(\psi(t) - \theta_i(t)))\mathrm{d}t$

```
35      def update(frame):
36          nonlocal phi, omega, K
37          x = np.cos(phi)
38          y = np.sin(phi)
39          xb, yb = np.mean(np.cos(phi)), np.mean(np.sin(phi))
40          psi = atan2(yb, xb)
41          p = sqrt(xb**2 + yb**2)
42          phi += (omega + K * p * np.sin(psi - phi)) * dt
43          points.set_data(x, y)
44          bari.set_data(xb, yb)
45          return points, bari,
```

- Initialisation des phases et des fréquences :

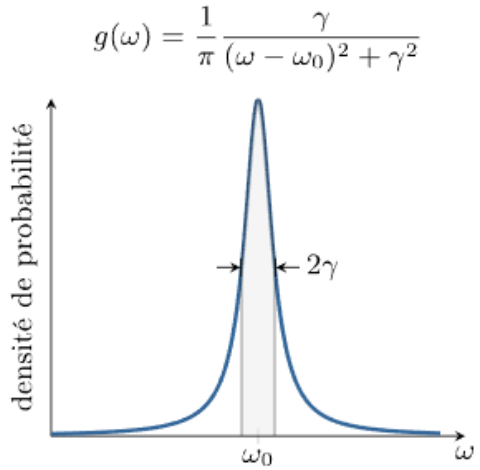


Figure 2 – Distribution lorentzienne des ω_i

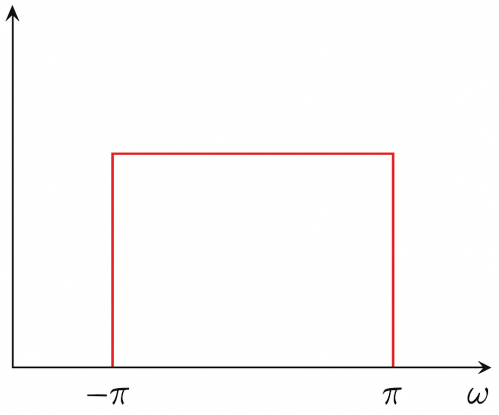


Figure 3 – Distribution uniforme des φ_i^0

- On choisit $\omega_0 = 1$ et $\mathrm{d}t = 0.01$

- Résultat pour $N = 50$, $\gamma = 0.5$ et $K = 2$:

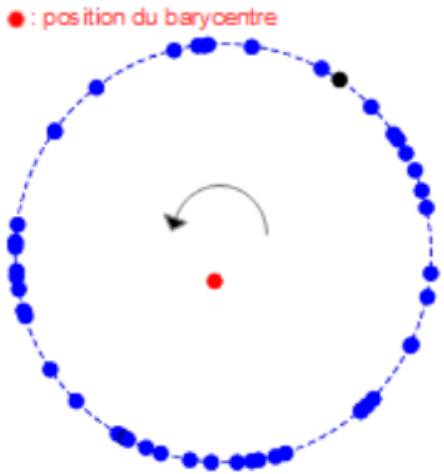


Figure 4 – État initial

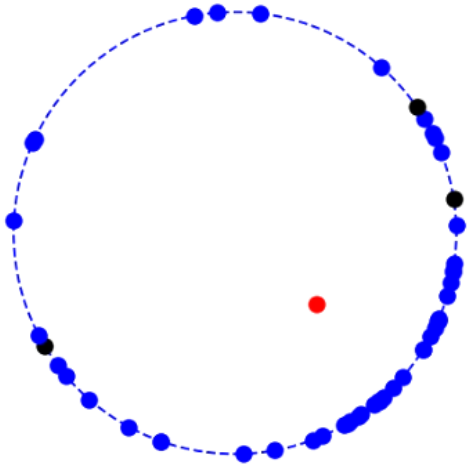


Figure 5 – État intermédiaire

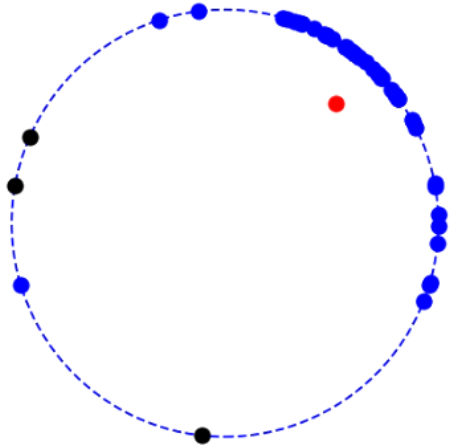


Figure 6 – État final

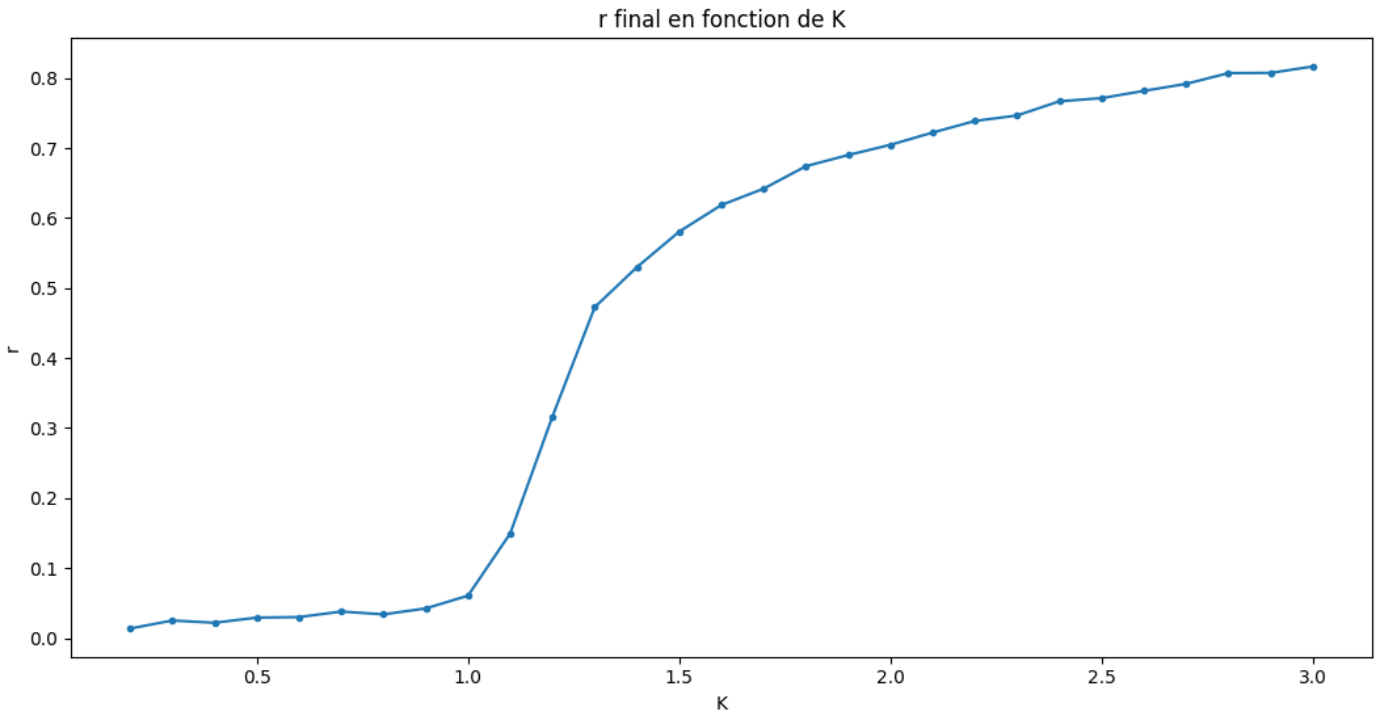


Figure 7 – Cas de $N = 3000$, $\gamma = 0.5$, $\mathrm{d}t = 0.01$

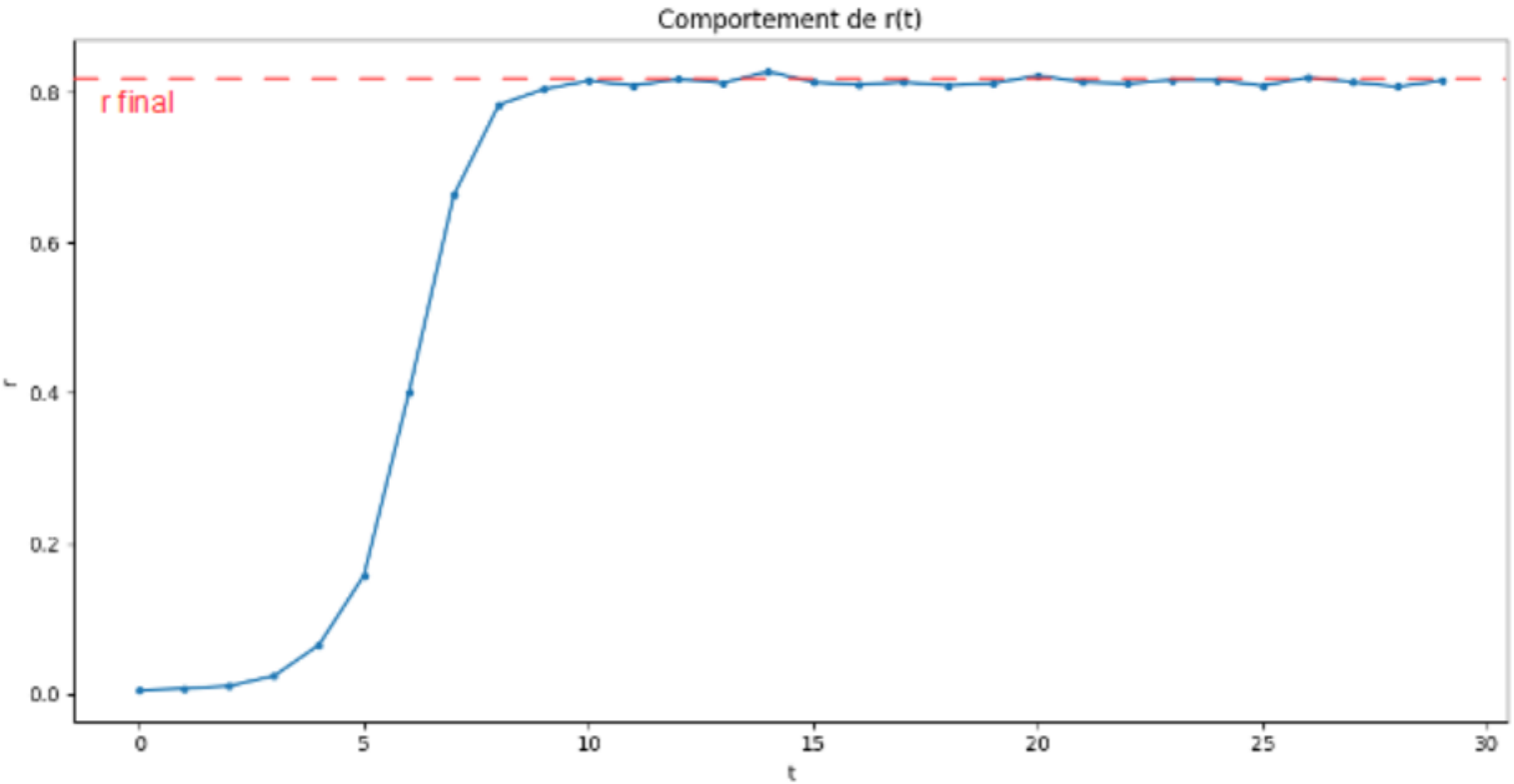


Figure 8 – Cas de $N = 3000$, $\gamma = 0.5$, $K = 3$, $\mathrm{d}t = 0.01$

- On change de référentiel tel que $\psi = 0$
- On choisit $\omega_0 = 0$ pour que $g(\omega) = g(-\omega)$

$$\Longrightarrow \boxed{\forall i \in [\![1, N]\!],\ \dot{\theta}_i(t) = \omega_i - K r(t) \sin(\theta_i(t))}$$

- Hypothèse : il existe 1 solution d'équilibre stable : r_∞=cste $\neq 0$
- On distingue 2 types d'oscillateurs :

  - les oscillateurs "locked" **L** : $|\omega_i| \leq K r_\infty$ donc $\dot{\theta}_i \to 0$

  - les oscillateurs "drifted" **D** : $|\omega_i| > K r_\infty$ donc $\dot{\theta}_i < 0$ ou $\dot{\theta}_i > 0$

- Probabilité de présence de l'oscillateur **D** de fréquence ω entre θ et $\theta + \mathrm{d}\theta$:

$$\rho(\theta, \omega)\mathrm{d}\theta$$

- Hypothèse de stationnarité :

$$\rho(\theta, \omega) \left| \frac{\mathrm{d}\theta}{\mathrm{d}t} \right| = C = \text{cste} > 0$$

$$\Rightarrow \rho(\theta, \omega) = \frac{C}{\left|\dot{\theta}\right|} = \frac{C}{|\omega - Kr \sin\theta|}$$

- Normalisation de la loi de probabilité (cf Annexe 1 page 24) :

$$\int_{-\pi}^{\pi} \rho(\theta,\omega)\,\mathrm{d}\theta = 1 \Rightarrow C\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = 1$$

$$\text{On pose : } \theta = 2Arctan(x)$$

$$\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = 2\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{|\omega + \omega x^2 - 2Krx|}$$

$$\Delta = 4K^2r^2 - 4\omega^2 = 4(K^2r^2 - \omega^2) < 0 \,\text{donc}\, \left|\omega + \omega x^2 - 2Krx\right| > 0$$

$$\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = \frac{2}{\omega}\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\left(x - \frac{Kr}{\omega}\right)^2 + \left(\frac{\sqrt{-\Delta}}{\omega}\right)^2} = \frac{2\pi}{\sqrt{\omega^2 - K^2r^2}}$$

$$\Longrightarrow \boxed{C = \frac{\sqrt{\omega^2 - K^2r^2}}{2\pi}} \text{ et } \boxed{\rho(\theta,\omega) = \frac{\sqrt{\omega^2 - K^2r^2}}{2\pi\,|\omega - Kr\sin\theta|}}$$

- On calcule r en étudiant le comportement de la totalité des oscillateurs (champ moyen) :

$$\psi = 0 \text{ donc } r = r\mathrm{e}^{\mathrm{i}\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{\mathrm{i}\theta_j} = < \mathrm{e}^{\mathrm{i}\theta} >$$

$$\text{Ainsi, } r = < \mathrm{e}^{\mathrm{i}\theta} >_L + < \mathrm{e}^{\mathrm{i}\theta} >_D$$

- Cas des oscillateurs "drifted" (cf Annexe 2 page 25) :

$$< \mathrm{e}^{\mathrm{i}\theta} >_D = \int_{-\pi}^{\pi} \int_{|\omega|>Kr} \mathrm{e}^{\mathrm{i}\theta} \rho(\theta,\omega) g(\omega)\, \mathrm{d}\omega\, \mathrm{d}\theta$$

$$\text{or, } g(\omega) = g(-\omega)$$

$$\text{et } \rho(\theta,\omega) = \frac{\sqrt{\omega^2 - K^2 r^2}}{2\pi\,|\omega - Kr sin\theta|} \text{ donc } \rho(\theta+\pi,-\omega) = \rho(\theta,\omega)$$

$$\text{Ainsi, } < \mathrm{e}^{\mathrm{i}\theta} >_D = 0$$

- Cas des oscillateurs "locked" :

$$< \mathrm{e}^{\mathrm{i}\theta} >_L = < \cos\theta >_L + \mathrm{i} < \sin\theta >_L$$

$$\text{or, } |\omega| \leq Kr \Longrightarrow \dot{\theta} \to 0 \Longrightarrow \sin(\theta) \to \frac{\omega}{Kr}$$

$$\text{Donc } < \sin\,\theta\, >_L = < \frac{\omega}{Kr_\infty} > = 0 \text{ car } g(\omega) = g(-\omega)$$

$$< \mathrm{e}^{\mathrm{i}\theta} >_L = < \cos\theta >_L + \mathrm{i} < \sin\theta >_L = < \cos\theta >_L = \int_{-Kr}^{Kr} \cos(\theta(\omega)) g(\omega)\, \mathrm{d}\omega$$

$$\text{on pose } \omega = Kr\sin\theta \text{ donc } \mathrm{d}\omega = Kr\cos\theta \mathrm{d}\theta$$

$$< \mathrm{e}^{\mathrm{i}\theta} >_L = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(\theta) g(Kr\sin\theta) Kr\cos\theta\, \mathrm{d}\theta = Kr \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta) g(Kr\sin\theta)\, \mathrm{d}\theta}_{F_K(r)}$$

Ainsi, $\boxed{r_\infty = Kr_\infty F_K(r_\infty)}$ avec $\boxed{F_K(r) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta) g(Kr\sin\theta)\, \mathrm{d}\theta}$

Il y a 2 solutions au problème :

- Ou bien $r_\infty = 0$: pas de transition vers un état synchronisé
- Ou bien $r_\infty > 0$: transition vers un état synchronisé : $\boxed{F_K(r_\infty) = \frac{1}{K}}$

- Calcul de $F_K(r)$ pour r>0 (cf Annexe 3 page 26) :

$$F_K(r) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta) \frac{\gamma}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta$$

$$F_K(r) = \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta}_{A} - \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma \sin^2\theta}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta}_{B}$$

$$B = \frac{\gamma - \gamma^2 A}{K^2 r^2} \text{ et } A = \frac{1}{\sqrt{\gamma^2 + K^2 r^2}}$$

Ainsi, $$\boxed{F_K(r) = \frac{\gamma}{K^2 r^2}\left(\sqrt{1 + \frac{K^2 r^2}{\gamma^2}} - 1\right)}$$

$$\text{En } 0,\ F_K(r) \sim \frac{1}{2\gamma} = \frac{1}{Kc} \text{ donc } \boxed{K_c = 2\gamma}$$

- Visualisation pour $\gamma = 0.5$ donc $K_c = 1$:

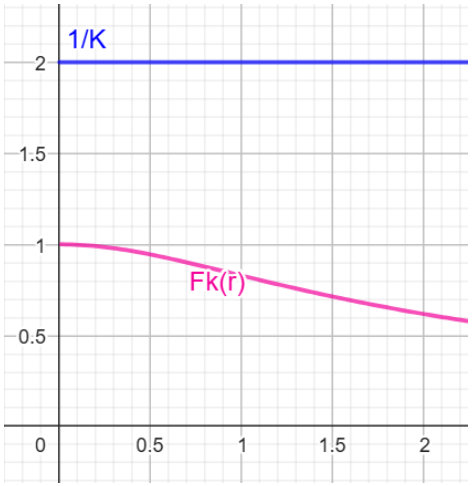


Figure 9 – $K = 0.5 < K_c$

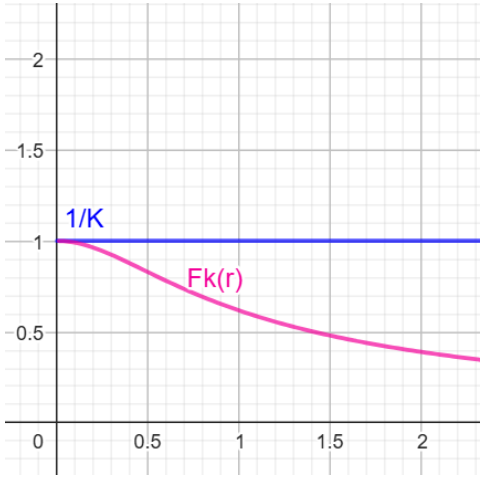


Figure 10 – $K = 1 = K_c$

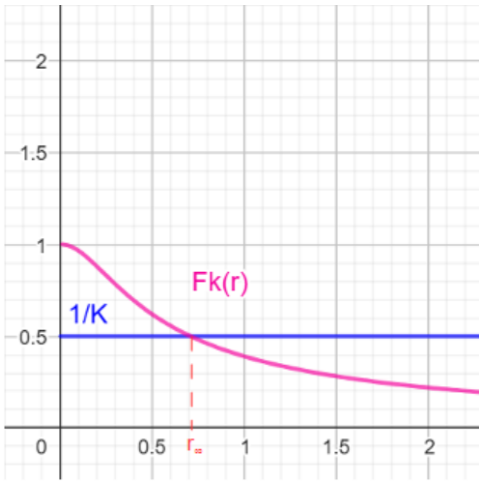


Figure 11 – $K = 2 > K_c$

Calcul de r_∞ (cf Annexe 4 page 27) :

$$F_K(r) = \frac{\gamma}{K^2 r^2}\left(\sqrt{1+\frac{K^2r^2}{\gamma^2}}-1\right) \text{ on pose } \gamma = \frac{K_c}{2}$$

$$F_K(r_\infty) = \frac{1}{K} \iff \underbrace{4\frac{Kr_\infty^2}{K_c}}_{\neq 0}\left(\frac{K}{K_c}-\frac{Kr_\infty^2}{K_c}-1\right) = 0$$

$$\implies r_\infty^2 = \frac{K_c}{K}\left(1-\frac{K}{K_c}\right)$$

Ainsi, $\boxed{r_\infty = \sqrt{1-\frac{K_c}{K}}}$ pour $K > K_c$

4. Conclusion

Mise en cohérence avec la modélisation de K_C

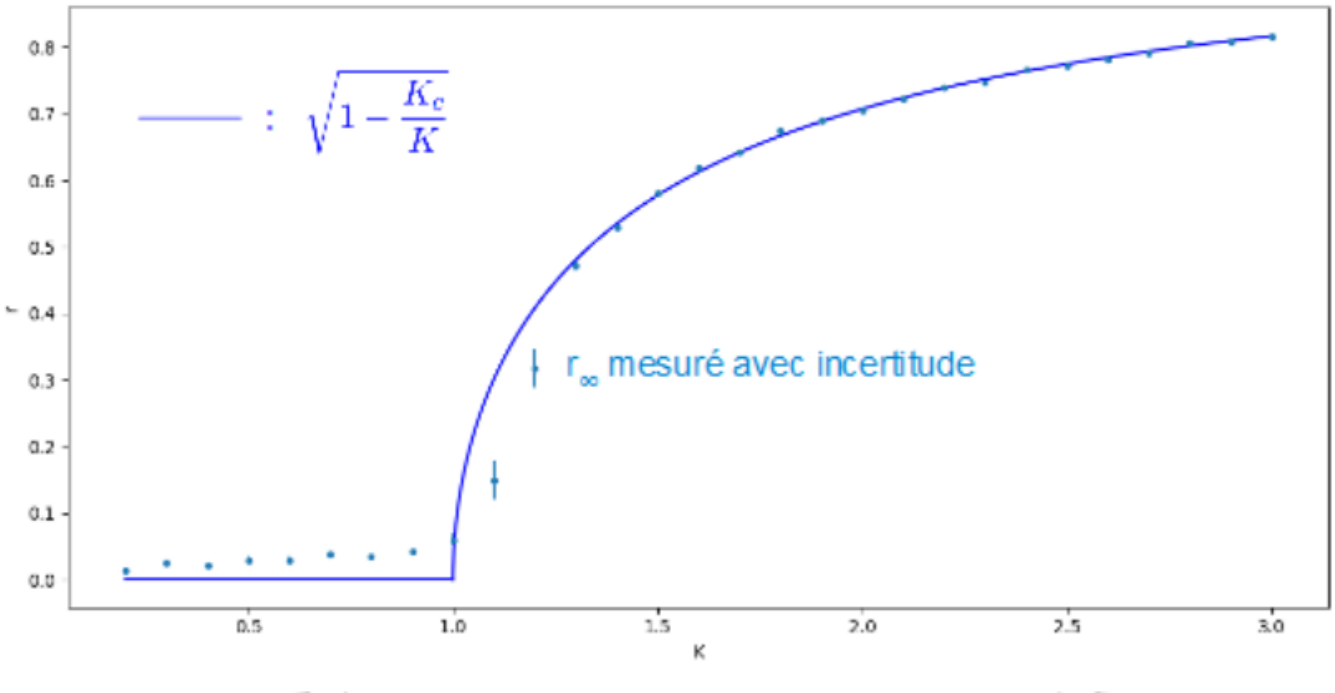


Figure 12 – Cas de N=3000, γ=0.5, dt=0.01

4. Conclusion

Mise en cohérence avec la modélisation de r_∞

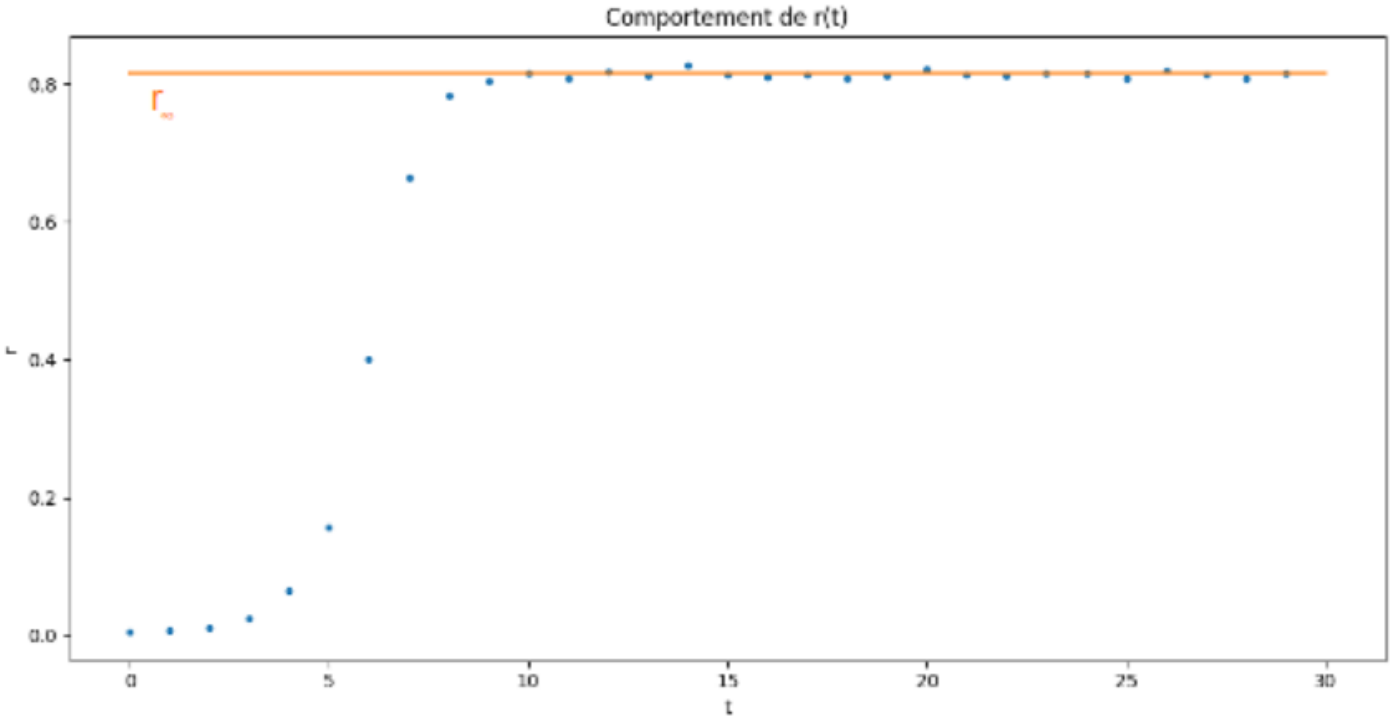


Figure 13 – Cas de N=3000, γ=0.5 , K=3, dt=0.01



Figure 14 – État désynchronisé

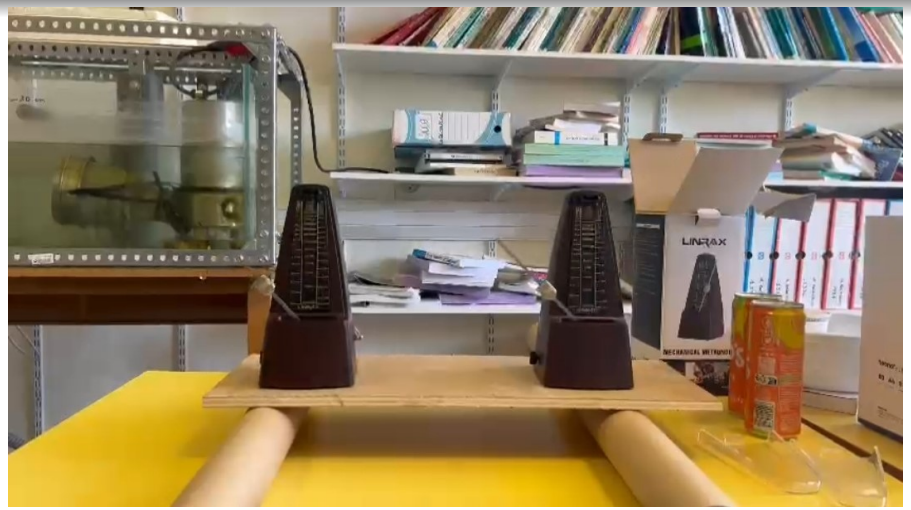


Figure 15 – État synchronisé

- Equations couplées :

$$\frac{\mathrm{d}^2\theta_1}{\mathrm{d}t^2} + \omega_1^2 \sin\theta_1 + \epsilon\left(\left(\frac{\theta_1}{\theta_0}\right)^2 - 1\right)\frac{\mathrm{d}\theta_1}{\mathrm{d}t} - \beta\cos\theta_1 \frac{\mathrm{d}^2}{\mathrm{d}t^2}(\sin\theta_1 + \sin\theta_2) = 0$$

$$\frac{\mathrm{d}^2\theta_2}{\mathrm{d}t^2} + \omega_2^2 \sin\theta_2 + \epsilon\left(\left(\frac{\theta_2}{\theta_0}\right)^2 - 1\right)\frac{\mathrm{d}\theta_2}{\mathrm{d}t} - \beta\cos\theta_2 \frac{\mathrm{d}^2}{\mathrm{d}t^2}(\sin\theta_1 + \sin\theta_2) = 0$$

- Condition de synchronisation :

$$\left|\frac{\omega_2 - \omega_1}{2}\right| \leq \frac{\sqrt{2}}{8}(\theta_0^2 + 2\beta)$$

Calcul complet de C :

$$\int_{-\pi}^{\pi} \rho(\theta,\omega)\,\mathrm{d}\theta = 1 \Rightarrow C\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = 1$$

$$\text{On pose : } \theta = 2Arctan(x)\ ,\ \mathrm{d}\theta = \frac{2\mathrm{d}x}{1+x^2}\ ,\ \sin\theta = \frac{2\tan(\frac{\theta}{2})}{1+\tan^2(\frac{\theta}{2})}$$

$$\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = 2\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{(1+x^2)\left|\omega - Kr\frac{2x}{1+x^2}\right|} = 2\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{|\omega + \omega x^2 - 2Krx|}$$

$$\Delta = 4K^2r^2 - 4\omega^2 = 4(K^2r^2 - \omega^2) < 0 \text{ donc } \left|\omega + \omega x^2 - 2Krx\right| > 0$$

$$\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = 2\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\omega\left(\left(x - \frac{Kr}{\omega}\right)^2 + 1 - \frac{K^2r^2}{\omega^2}\right)} = \frac{2}{\omega}\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\left(x - \frac{Kr}{\omega}\right)^2 + \left(\frac{\sqrt{-\Delta}}{\omega}\right)^2}$$

$$\int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{|\omega - Kr\sin\theta|} = \frac{2}{\omega}\left[\frac{\omega}{\sqrt{-\Delta}}\arctan\left(\frac{x - \frac{Kr}{\omega}}{\frac{\sqrt{-\Delta}}{\omega}}\right)\right]_{-\infty}^{\infty} = \frac{2\pi}{\sqrt{\omega^2 - K^2r^2}}$$

$$\Rightarrow C = \frac{\sqrt{\omega^2 - K^2r^2}}{2\pi} \text{ et } \rho(\theta,\omega) = \frac{\sqrt{\omega^2 - K^2r^2}}{2\pi\,|\omega - Kr\sin\theta|}$$

Calcul de $< \mathrm{e}^{\mathrm{i}\theta} >_D$:

$$< \mathrm{e}^{\mathrm{i}\theta} >_D = \int_{-\pi}^{\pi} \int_{|\omega|>Kr} \mathrm{e}^{\mathrm{i}\theta} \rho(\theta,\omega) g(\omega)\, \mathrm{d}\omega\, \mathrm{d}\theta = I$$

$$I = \underbrace{\int_{-\pi}^{\pi} \int_{\omega>Kr} \mathrm{e}^{\mathrm{i}\theta} \rho(\theta,\omega) g(\omega)\, \mathrm{d}\omega\, \mathrm{d}\theta}_{I_+} + \underbrace{\int_{-\pi}^{\pi} \int_{\omega<Kr} \mathrm{e}^{\mathrm{i}\theta} \rho(\theta,\omega) g(\omega)\, \mathrm{d}\omega\, \mathrm{d}\theta}_{I_-}$$

Pour I_- , on pose : $\theta' = \theta - \pi$, $\omega' = -\omega$

$$I_- = \int_0^{2\pi} - \int_{\omega'>Kr} \mathrm{e}^{\mathrm{i}(\theta'+\pi)} \rho(\theta'+\pi, -\omega') g(-\omega')\, (-\mathrm{d}\omega')\, \mathrm{d}\theta'$$

or, $\rho(\theta+\pi, -\omega) = \rho(\theta,\omega)$ et $g(-\omega) = g(\omega)$

$$\text{Donc } I_- = -\int_0^{2\pi} \int_{\omega>Kr} \mathrm{e}^{\mathrm{i}\theta} \rho(\theta,\omega) g(\omega)\, \mathrm{d}\omega\, \mathrm{d}\theta$$

Ainsi, par 2π-périodicité de ρ pour θ et de $\theta \mapsto \mathrm{e}^{\mathrm{i}\theta}$, $I_- = -I_+$ donc $I = I_+ + I_- = 0$

Calcul complet de $F_K(r)$:

$$F_K(r) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta) \frac{\gamma}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta$$

$$F_K(r) = \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta}_{A} - \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma \sin^2\theta}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta}_{B}$$

$$\text{or, } \pi = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma^2 + K^2 r^2 \sin^2\theta}{\gamma^2 + K^2 r^2 \sin^2\theta} \mathrm{d}\theta = \pi\gamma A + K^2 r^2 \frac{\pi}{\gamma} B \text{ donc } B = \frac{\gamma - \gamma^2 A}{K^2 r^2}$$

$$A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\gamma}{\pi(\gamma^2 + K^2 r^2 \sin^2\theta)} \mathrm{d}\theta = \frac{1}{\pi\gamma} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\mathrm{d}\theta}{1 + \left(\frac{Kr}{\gamma}\right)^2 \frac{\tan^2\theta}{1+\tan^2\theta}} \text{ on pose } \tan\theta = x$$

$$A = \frac{1}{\pi\gamma} \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1 + x^2 + \left(\frac{Kr}{\gamma}\right)^2 x^2} = \frac{1}{\pi\gamma} \left[\frac{1}{\sqrt{1 + \left(\frac{Kr}{\gamma}\right)^2}} \arctan\left(x \sqrt{1 + \left(\frac{Kr}{\gamma}\right)^2} \right) \right]_{-\infty}^{\infty}$$

$$\text{Donc } A = \frac{1}{\sqrt{\gamma^2 + K^2 r^2}} \text{ donc, } F_K(r) = A - B = \frac{1}{\sqrt{\gamma^2 + K^2 r^2}} \left(1 + \frac{\gamma^2}{K^2 r^2}\right) - \frac{\gamma}{K^2 r^2}$$

$$\text{Ainsi, } F_K(r) = \frac{\gamma}{K^2 r^2} \left(\sqrt{1 + \frac{K^2 r^2}{\gamma^2}} - 1 \right)$$

Calcul complet de r_∞ :

$$F_K(r) = \frac{\gamma}{K^2 r^2}\left(\sqrt{1+\frac{K^2r^2}{\gamma^2}}-1\right) \text{ on pose } \gamma = \frac{K_c}{2}$$

$$\frac{K_c}{2K^2r_\infty^2}\left(\sqrt{1+\frac{4K^2r_\infty^2}{K_c^2}}-1\right) = \frac{1}{K}$$

$$\iff \frac{K_c}{2Kr_\infty^2}\left(\sqrt{1+\frac{4K^2r_\infty^2}{K_c^2}}-1\right) = 1 \iff \sqrt{1+\frac{4K^2r_\infty^2}{K_c^2}} = \frac{2Kr_\infty^2}{K_c}+1$$

$$\iff 1+\frac{4K^2r_\infty^2}{K_c^2} = \frac{4K^2r_\infty^4}{K_c^2}+\frac{4Kr_\infty^2}{K_c}+1 \iff \underbrace{4\frac{Kr_\infty^2}{K_c}}_{\neq 0}\left(\frac{K}{K_c}-\frac{Kr_\infty^2}{K_c}-1\right) = 0$$

$$\implies \frac{K}{K_c}-\frac{Kr_\infty^2}{K_c}-1 = 0 \iff r_\infty^2 = \frac{K_c}{K}\left(1-\frac{K}{K_c}\right)$$

$$\text{Ainsi, } r_\infty = \sqrt{1-\frac{K_c}{K}} \quad \text{pour } K > K_c$$

Code pour l'animation :

```
fig, ax = plt.subplots()
ax.set_aspect('equal')
ax.set_xlim(-1.5, 1.5)
ax.set_ylim(-1.5, 1.5)
ax.set_axis_off()
circle = plt.Circle((0, 0), 1, fill=False, color='blue', linestyle='--')
ax.add_artist(circle)
points, = ax.plot([], [], 'ro')
bari, = ax.plot([], [], 'bo')

def test(N, Gamma, K):
    w0 = 1
    dt = 0.01
    duree = 2000 * dt
    phi = np.array([uniform(-np.pi, np.pi) for _ in range(N)])
    omega = np.array(cauchy.rvs(loc=w0, scale=Gamma, size=N))

    def init():
        points.set_data([], [])
        bari.set_data([], [])
        return points, bari,

    def update(frame):
        nonlocal phi, omega, K
        x = np.cos(phi)
        y = np.sin(phi)
        xb, yb = np.mean(np.cos(phi)), np.mean(np.sin(phi))
        psi = atan2(yb, xb)
        p = sqrt(xb**2 + yb**2)
        phi += (omega + K * p * np.sin(psi - phi)) * dt
        points.set_data(x, y)
        bari.set_data(xb, yb)
        return points, bari,

    ani = FuncAnimation(fig, update, frames=int(duree / dt), init_func=init, interval=1000 * dt)
    plt.show()


test(N=30, Gamma=0.1, K=1.5)

```

Code pour trouver K_C :

```
P=[]
lp = []
lk = []
perr=[]


def test(N, Gamma, K, c):
    w0 = 1
    global phi, omega
    phi = np.array([uniform(-np.pi, np.pi) for _ in range(N)])
    omega = np.array(cauchy.rvs(loc=w0, scale=Gamma, size=N))
    for _ in range(3000):
            xb, yb = np.mean(np.cos(phi)), np.mean(np.sin(phi))
            psi = atan2(yb, xb)
            p = sqrt(xb**2 + yb**2)
            phi += (omega + K * p * np.sin(psi - phi)) * 0.01
    if c<10:
        c+=1
        P.append(p)
        test(N, Gamma,K,c)
    elif K > 0.1:
        lp.append(np.mean(P))
        perr.append(np.std(P)/np.sqrt(len(P)))
        lk.append(K)
        P.clear()
        print(K)
        test(N, Gamma, round(K - 0.1, 2),1)
    else:
        display_graph()

def display_graph():
    fig_graph, ax_graph = plt.subplots()
    ax_graph.set_xlabel('K')
    ax_graph.set_ylabel('r')
    ax_graph.set_title('Comportement de r en fonction de K')
    ax_graph.errorbar(lk, lp, yerr=perr,fmt='.')
    x1 = np.linspace(0.2, 1, 200)
    y1 = np.zeros_like(x1)
    x2 = np.linspace(1, 3, 900)
    y2 = np.sqrt(1 - 1 / x2)

    plt.plot(x2, y2, color='blue')
    plt.plot(x1, y1, color='blue')
    plt.show()


test(N=3000, Gamma=0.5, K=3, c=1)
```

Code pour trouver r_∞ :

```
lp = []
lt = []

def test(N, Gamma, K):
    w0 = 1
    global phi, omega
    phi = np.array([uniform(-np.pi, np.pi) for _ in range(N)])
    omega = np.array(cauchy.rvs(loc=w0, scale=Gamma, size=N))
    for s in range(3000):
            xb, yb = np.mean(np.cos(phi)), np.mean(np.sin(phi))
            psi = atan2(yb, xb)
            p = sqrt(xb**2 + yb**2)
            phi += (omega + K * p * np.sin(psi - phi)) * 0.01
            if s%100==0:
                lp.append(p)
                lt.append(s//100)
    display_graph()


def display_graph():
    fig_graph, ax_graph = plt.subplots()
    ax_graph.set_xlabel('t')
    ax_graph.set_ylabel('r')
    ax_graph.set_title('Comportement de r en fonction de t')
    ax_graph.plot(lt, lp)
    plt.show()


test(N=3000, Gamma=0.5, K=3)

```